

Adopted Levels, Gammas

$Q(\beta^-)=167.322\ 26$; $S(n)=6985.84\ 4$; $S(p)=11586.5\ 8$; $Q(\alpha)=-8322.09\ 6$ [2021Wa16](#)

$S(2n)=18402.99\ 4$, $S(2p)=22909.8\ 7$ ([2021Wa16](#)).

Isotope discovery ([2012Th10](#)): $\text{Cl}(n,X)$ reaction at Physical Institute of Aarhus University ([1936An01](#)).

^{35}S decay measurements:

[2014Si21](#), [2000Ho13](#), [1999Pa18](#), [1999Oh09](#).

^{35}S cross-section measurements:

$^{35}\text{Cl}(n,p)^{35}\text{S}$: [2024Ha18](#), [2023De34](#), [2021WaZY](#), [2020Ku17](#), [2019Ba16](#), [2000Ka30](#), [1999Gl09](#), [1999Gl02](#);

Projectile fragmentation $^9\text{Be}(^{40}\text{Ar},X)$ and $^{181}\text{Ta}(^{40}\text{Ar},X)$: [2012Zh06](#), [2007No13](#);

Spallation $^{136}\text{Xe}(p,X)$: [2007Na31](#);

$^{34}\text{S}(n,\gamma)$ $E=\text{thermal}$: [2023KoZS](#), [2008FiZZ](#).

 ^{35}S LevelsCross Reference (XREF) Flags

A	$^{35}\text{P}\ \beta^-$ decay (47.3 s)	F	$^{34}\text{S}(n,\gamma)$ $E=\text{thermal}$	K	$^{37}\text{Cl}(p,^3\text{He})$
B	$^2\text{H}(^{34}\text{S},p\gamma)$	G	$^{34}\text{S}(n,n),(n,\gamma)$: resonances	L	$^{37}\text{Cl}(d,\alpha\gamma)$
C	$^9\text{Be}(^{36}\text{S},^{35}\text{S}\gamma)$	H	$^{34}\text{S}(\text{pol } d,p)$	M	$^{160}\text{Gd}(^{34}\text{S},x\gamma),(^{37}\text{Cl},X\gamma)$
D	$^{24}\text{Mg}(^{14}\text{N},3p\gamma)$	I	$^{34}\text{S}(d,p)$	N	$^{208}\text{Pb}(^{36}\text{S},^{35}\text{S}\gamma)$
E	$^{26}\text{Mg}(^{18}\text{O},2\alpha n\gamma)$	J	$^{34}\text{S}(d,p\gamma)$		

$E(\text{level})^\dagger$	$J^\pi^\ddagger$	$T_{1/2}^\#$	XREF	Comments
0.0 ^{&}	$3/2^+$	87.61 d 24	ABCDEF HIJKLMN	$\% \beta^- = 100$ $\mu = (+)1.00\ 4$ (1954Bu05 , 2019StZV) $Q = 0.0483\ 3$ (2018Py01 , 2021StZZ) μ : $+1.00\ 4$ or $-1.07\ 4$ from 1954Bu05 using the microwave absorption method, and the negative value is less likely from systematics of odd-A S isotopes. Q : from 2018Py01 using the microwave absorption method. $-0.483\ 3$ is given in 2021StZZ . Other: 0.045 from 1954Bi40 using the microwave absorption method. J^π : $L=2$ from 0^+ in (pol d,p), (d,p), and $^9\text{Be}(^{36}\text{S},^{35}\text{S})$ and $L-1/2$ transfer from analyzing power. $J=3/2$ also from microwave spectroscopy (1949Co12 , 1951We11). $T_{1/2}$: unweighted average of 88 d 3 (1941Ka01), 87.1 d 12 (1947Hu06), 86.35 d 17 (1959Co56), 87.16 d 10 (1958Se49), 88.8 d 10 (1959Ca12), 89.0 d 5 (1961Wy01), 87.1 d 9 (1961Oz01), 87.9 d 3 (1965Fl02), 87.39 d 10 (1968Wo06), 87.5 d 4 (1969La34), and 87.38 d 3 (1999Pa18).
1572.378 19	$1/2^+$	2.29 ps 14	ABCD F HIJKL N	J^π : $L=0$ from 0^+ in (pol d,p), (d,p), and $^9\text{Be}(^{36}\text{S},^{35}\text{S})$. $T_{1/2}$: from lifetime=3.3 ps 2 in $^{208}\text{Pb}(^{36}\text{S},^{35}\text{S}\gamma)$ from 2022Gr07 with DRDM. Others: lifetime=3.3 ps 5 in $(^{34}\text{S},p\gamma)$ from 1973Wa10 with DSAM; lifetime>1.8 ps in (d,p γ) from 1970Bu18 with DSAM; lifetime>1.6 ps in (d,p γ) from 1972Fr11 with DSAM.
1991.28 [@] 5	$7/2^-$	1.03 ns 5	CDEF HIJKLMN	J^π : $L(\text{pol } d,p)=3$ from 0^+ and $L+1/2$ transfer from analyzing power. $T_{1/2}$: from lifetime=1.48 ns 7: weighted average of 1.7 ns 3 from (d,p γ) from 2024Co04 with p γ -delayed coin and 1.47 ns 7 from (d,p γ) from 1971Pr11 with p γ -delayed coin. Other: >4.5 ps in (d,p γ) (1970Bu18 , DSAM).
2347.789 18	$3/2^-$	0.78 ps 12	BCD F HIJ L N	J^π : $L=1$ from 0^+ in (pol d,p), (d,p), and $^9\text{Be}(^{36}\text{S},^{35}\text{S})$ and $L+1/2$ transfer from analyzing power. $T_{1/2}$: from lifetime=1.12 ps 18: weighted average of 1.3 ps 2 in $(^{34}\text{S},p\gamma)$ from 1973Wa10 with DSAM, 0.8 ps 2 in (d,p γ) from 1970Bu18 with DSAM, and 1.22 ps 18 in (d,p γ) from 1972Fr11 with DSAM.

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Adopted Levels, Gammas (continued) ^{35}S Levels (continued)

E(level) [†]	J ^π [‡]	T _{1/2} [#]	XREF	Comments
2717.00 3	5/2 ⁺	69 fs 24	CD F HIJKL N	J ^π : L=2 from 0 ⁺ in (pol d,p), (d,p), and ⁹ Be(³⁶ S, ³⁵ S) and L+1/2 transfer from analyzing power. T _{1/2} : from lifetime=100 fs 35 in (d,pγ) from 1972Fr11 with DSAM.
2938.64 5	(3/2) ⁺		A C F HIJKL	J ^π : L=2 from 0 ⁺ in (d,p) and (pol d,p); J=3/2, 5/2 from γ(θ) in (d,pγ) (1972Va07,1975VaYG); Possibly allowed β ⁻ feeding from 1/2 ⁺ parent with log ft=5.0. Other: L=0 from 0 ⁺ in ⁹ Be(³⁶ S, ³⁵ S) is inconsistent.
3421.1 3	5/2 ⁺	<69 fs	C HIJKL N	J ^π : L=2 from 0 ⁺ in (pol d,p), (d,p), and ⁹ Be(³⁶ S, ³⁵ S) and L+1/2 transfer from analyzing power. T _{1/2} : from lifetime<100 fs in (d,pγ) from 1972Fr11 with DSAM.
3558.09 3	(3/2 ⁻ ,5/2 ⁺)		F HIJ L N	J ^π : 1986γ to 1/2 ⁺ and 1566.7γ to 7/2 ⁻ .
3594.8& 4	3/2 ⁺ ,5/2 ⁺		CDE IJKL N	XREF: C(3602)I(3596)K(3598) J ^π : L=2 from 0 ⁺ in ⁹ Be(³⁶ S, ³⁵ S).
3675? 10	(1/2 ⁻ ,3/2 ⁻)		I	J ^π : L=(1) from 0 ⁺ in (d,p).
3801.958 23	3/2 ⁻	25 fs 18	F HIJKL	XREF: k(3811) J ^π : L=1 from 0 ⁺ in (d,p) and (pol d,p) and L+1/2 transfer from analyzing power. T _{1/2} : from lifetime=36 fs 26 in (d,pγ) from 1972Fr11 with DSAM.
3818.1@ 4	5/2 ⁻ ,7/2 ⁻	0.28 ps 3	CDE H JkL N	XREF: k(3811) J ^π : L=3 from 0 ⁺ in ⁹ Be(³⁶ S, ³⁵ Sγ) and γ M1+E2 to 7/2 ⁻ . T _{1/2} : from (¹⁴ N,3pγ) in 2014Ay01 with DSAM.
3885.3 4	(3/2 ⁻ ,5/2 ⁺)		D IJ L	J ^π : 2312.6γ to 1/2 ⁺ and 1894.9γ to 7/2 ⁻ .
3895 3	1/2 ⁺		C	J ^π : L=0 from 0 ⁺ in ⁹ Be(³⁶ S, ³⁵ Sγ).
4022.3@ 10	11/2 ⁻	0.32 ps 3	DE MN	J ^π : 2032.0γ E2, ΔI=2 to 7/2 ⁻ . T _{1/2} : from (¹⁴ N,3pγ) in 2014Ay01 with DSAM.
4025.9 4	3/2 ⁺ ,5/2 ⁺		C IJKL	XREF: I(4025)K(4027) J ^π : L=2 from 0 ⁺ in ⁹ Be(³⁶ S, ³⁵ Sγ) and L=2 from 3/2 ⁺ in (p, ³ He); 2455.4γ to 1/2 ⁺ in (d,pγ) and (d,αγ).
4106.4 4	(5/2) ⁺	<55 fs	F IJKL	J ^π : 2535γ to 1/2 ⁺ and 2114γ to 7/2 ⁻ ; L=0+2 from 3/2 ⁺ in (p, ³ He). T _{1/2} : from lifetime<80 fs in (d,pγ) from 1972Fr11 with DSAM.
4180 3	(1/2,3/2,5/2 ⁺)		L	J ^π : γ to 1/2 ⁺ .
4189.280 21	1/2 ⁻	<35 fs	F HIJKL	T _{1/2} : from lifetime<50 fs in (d,pγ) from 1972Fr11 with DSAM. J ^π : L=1 from 0 ⁺ in (d,p) and (pol d,p) and L-1/2 transfer from analyzing power.
4302.6 7	(1/2 to 7/2 ⁻)		I L	J ^π : γ's to 3/2 ⁺ and 3/2 ⁻ .
4477.64 7	(1/2,3/2,5/2) ⁺	<62 fs	F IJKL	XREF: i(4481)K(4489) T _{1/2} : from lifetime<90 fs in (d,pγ) from 1972Fr11 with DSAM. J ^π : γ's to 1/2 ⁺ and L(p, ³ He)=2; J ^π =7/2 ⁻ and L=3 from (pol d,p) presumably erroneous.
4482	7/2 ⁻		Hi	XREF: i(4481)
4576 8	(1/2 to 5/2) ⁺		HI K	E(level): weighted average of 4575 8 from (d,p) and 4577 10 from (p, ³ He). J ^π : L(p, ³ He)=0+2???
4617 10			K	
4839 8	(1/2 to 9/2) ⁺		DE HI K	J ^π : L(p, ³ He)=2.
4899.7 4	9/2 ⁺		DE	

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Adopted Levels, Gammas (continued) ^{35}S Levels (continued)

E(level) [†]	J ^π [‡]	T _{1/2} [#]	XREF	Comments
4903.32 8	1/2 ⁻		F HI	J ^π : L=1 from 0 ⁺ in (d,p) and (pol d,p) and L-1/2 transfer from analyzing power.
4963.09 3	3/2 ⁻		F HI K	J ^π : L=1 from 0 ⁺ in (d,p) and (pol d,p) and L+1/2 transfer from analyzing power.
4990 10	(1/2 to 5/2) ⁺		K	J ^π : L(p, ³ He)=0+2???
5010.2 3	11/2 ⁻	0.45 ps 8	DE	T _{1/2} : from (¹⁴ N,3pγ) in 2014Ay01 with DSAM.
5058 8	7/2 ⁻		HI	E(level): from (d,p). J ^π : L=3 from 0 ⁺ in (d,p) and (pol d,p) and L+1/2 transfer from analyzing power.
5127 10	(1/2 to 9/2) ⁺		HI K	E(level): from (p, ³ He). Other: 5126 11 from (d,p). J ^π : L(p, ³ He)=2???
5343 8	(1/2 to 11/2) ⁻		I K	E(level): weighted average of 5342 8 from (d,p) and 5345 10 from (p, ³ He). J ^π : L(p, ³ He)=3???
5412.2 3	9/2 ⁺		DE	XREF: I(?)
5475? 10			I	XREF: I(5542?)
5550 10	(1/2 to 11/2) ⁽⁻⁾		I K	J ^π : L(p, ³ He)=(3)???
5752.5 8	(1/2 to 9/2) ⁻		F I k	XREF: I(5740)k(5771) J ^π : γ to 3/2 ⁺ .
5771 10	(1/2 to 9/2) ⁺		k	XREF: k(5771) J ^π : L(p, ³ He)=2.
5841.485 31			F	
5877.6& 3	11/2 ⁺		DE	
5890 20			I K	XREF: K(5915?)
5980 10			I	
6018.8 6	(1/2 to 9/2) ⁻		F	J ^π : γ to 3/2 ⁺ .
6078.48 4	1/2 ⁻ , 3/2 ⁻		F I	J ^π : L(d,p)=1.
6129 10	(1/2, 3/2, 5/2) ⁺		K	J ^π : L(p, ³ He)=0+2.
6293.93 6	(1/2 to 9/2) ⁻		F i	XREF: i(6292) J ^π : γ to 3/2 ⁺ .
6299.4 4	11/2 ⁺		D i	XREF: i(6292)
6334@ 8		50 fs 10	D I k	XREF: k(6347) T _{1/2} : from (¹⁴ N,3pγ) in 2014Ay01 with DSAM.
6354.89 23	(1/2 to 9/2) ⁻		F I k	XREF: k(6347) J ^π : γ to 3/2 ⁺ .
6419.9 11	(1/2 to 9/2) ⁻		F	J ^π : γ to 3/2 ⁺ .
6446 8			I	
6496 8			I	
6537.7 14			I	
6545.1 13			I	
6584 10			I	
6629.43 10	(1/2 to 9/2) ⁻		F	J ^π : γ to 3/2 ⁺ .
6635.2 13			i k	XREF: k(6654)
6654 10			i k	XREF: k(6654)
6684 9	(1/2 to 9/2) ⁺		I K	E(level): weighted average of 6677 8 from (d,p) and 6696 10 from (p, ³ He). J ^π : L(p, ³ He)=2.
6761.0 12	(1/2 to 9/2) ⁻		F	J ^π : γ to 3/2 ⁺ .
6891.3 14	(1/2 to 7/2) ⁽⁻⁾		I	
(6986.10 3)	1/2 ⁺		FG	XREF: G(?)
7018.89 4		0.025 eV 1	G I	J ^π : s-wave resonance capture state. E(level): from (n,n). Other: 7021 10 from (d,p). T _{1/2} : from (n,n).

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Adopted Levels, Gammas (continued) ^{35}S Levels (continued)

E(level) [†]	J ^π [‡]	T _{1/2} [#]	XREF	Comments
7072.66 5		0.14 eV 4	G	
7074.33 6		0.084 eV 3	G	
7097.72 30		0.009 eV 3	G i k	XREF: i(7150)k(7151)
7099.85 20		0.033 eV 5	G i k	XREF: i(7150)k(7151)
7100.73 4	3/2		G	J ^π : R-Matrix analysis of resonances in (n,n).
7143.27 11		0.42 eV 1	G i k	XREF: i(7150)k(7151)
7179.8 & 3	15/2 ⁺	3.1 ps 12	DE	T _{1/2} : from ($^{14}\text{N}, 3p\gamma$) in 2014Ay01 with DSAM.
7210.42 5			G i	XREF: i(7205)
7218.0 4		0.33 eV 2	G i	XREF: i(7205)
7234.0 5		0.19 eV 1	G i	XREF: i(7205)
7239.8 6		0.06 eV 2	G i	XREF: i(7205)
7253.2 6		0.35 eV 2	G i	XREF: i(7205)
7275.93 5	1/2 ⁺		G	J ^π : L(n,n)=0.
7279			G	XREF: G(?)
7289.8 8		0.31 eV 3	G	
7294.19 6	3/2 ⁻		G	J ^π : L(n,n)=1 and R-Matrix analysis of resonances in (n,n).
7331.00 5	1/2 ⁺		G	J ^π : L(n,n)=0.
7338.18 20			G	
7343.91 20		0.72 eV 3	G	
7347.2 5		0.44 eV 6	G	
7347.9 5		0.30 eV 5	G	
7353.9 5		0.44 eV 4	G k	XREF: k(7375)
7357.6 5		0.16 eV 4	G k	XREF: k(7375)
7370.56 6	1/2 ⁻		G k	XREF: k(7375)
7375 10			G k	J ^π : L(n,n)=1 and R-Matrix analysis of resonances in (n,n).
7396.0 6		0.44 eV 33	G k	XREF: G(?)k(7375)
7404.6 6		0.21 eV 5	G	XREF: k(7375)
7408.68 11	3/2 ⁻		G	J ^π : L(n,n)=1 and R-Matrix analysis of resonances in (n,n).
7411.55 6			G	
7416.50 4			G	
7429.1 6		0.12 eV 3	G	
7433.56 5			G	
7436.4 4			G	
7442.14 5	1/2 ⁺		G	J ^π : L(n,n)=0.
7462.2 6		0.33 eV 4	G	
7481.15 7	1/2 ⁻		G I	E(level): weighted average of 7481.15 6 from (n,n) and 7482.7 13 from (d,p).
7494.5 6		0.17 eV 4	G	J ^π : L(n,n)=1 and R-Matrix analysis of resonances in (n,n).
7542.3 6		0.21 eV 3	G i	XREF: i(7570)
7604.0 6		0.90 eV 8	G i	XREF: i(7570)
7609.26 5	3/2 ⁻		G	J ^π : L(n,n)=1 and R-Matrix analysis of resonances in (n,n).
7648.9 6		0.21 eV 3	G i	XREF: i(7640)
7655.7 6		0.80 eV 5	G i	XREF: i(7640)
7663.91 11			G i	XREF: i(7640)
7678.3 7		0.45 eV 7	G i	XREF: i(7640)
7712? 10	(3/2 to 13/2) ⁺		g k	XREF: k(7712)
7731.1 7			g k	J ^π : L(p, ^3He)=4.
7749.7 4			G k	XREF: k(7712)
7761.46 7	3/2 ⁻		G k	XREF: k(7770)
7776.17 11	1/2 ⁻		G k	J ^π : L(n,n)=1 and R-Matrix analysis of resonances in (n,n).
7797.99 9	1/2 ⁺		G	XREF: k(7770)
				J ^π : L(n,n)=1 and R-Matrix analysis of resonances in (n,n).
				J ^π : L(n,n)=0.

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Adopted Levels, Gammas (continued) ^{35}S Levels (continued)

E(level) [†]	J^π [‡]	$T_{1/2}$ [#]	XREF	Comments
7812.24 7	3/2 ⁺		G	J^π : L(n,n)=2 and R-Matrix analysis of resonances in (n,n).
7853.25 7	3/2 ⁻		G	J^π : L(n,n)=1 and R-Matrix analysis of resonances in (n,n).
7861.8 12		1.13 eV 19	G	
7880.8 12		0.67 eV 17	G	
7889.0 12		0.57 eV 25	G	
7894.63 7	3/2 ⁻		G	J^π : L(n,n)=1 and R-Matrix analysis of resonances in (n,n).
7899.70 30			G	
7934.1 15		0.5 eV 3	G	
7939.5 15		0.89 eV 4	G	
7954.96 11	3/2 ⁻		G	J^π : L(n,n)=1 and R-Matrix analysis of resonances in (n,n).
7974.19 11	3/2 ⁻		G	J^π : L(n,n)=1 and R-Matrix analysis of resonances in (n,n).
8019.5 2			G	
8023.7 ^{&} 9	17/2 ⁺	0.15 ps 4	DE	$T_{1/2}$: from ($^{14}\text{N}, 3p\gamma$) in 2014Ay01 with DSAM.
8041.20 20			G	
8076.46 20			G	
8077.91 30			G	
8093.0 5	1/2 ⁻		G K	E(level): from (n,n). Other: 8103 10 from (p, ^3He). J^π : L(n,n)=1 and R-Matrix analysis of resonances in (n,n).
8143.75 20			g k	XREF: k(8160)
8160 10	(1/2 to 7/2) ⁽⁻⁾		g k	XREF: k(8160) J^π : L(p, ^3He)=(1).
8187.17 30	1/2 ⁻		G	J^π : L(n,n)=1 and R-Matrix analysis of resonances in (n,n).
8211.25 30			G	
8224.2 4	1/2 ⁻		G	J^π : L(n,n)=1 and R-Matrix analysis of resonances in (n,n).
8228.3 4			G	
8243.98 30			G	
8256.31 20	3/2 ⁻		G	J^π : L(n,n)=1 and R-Matrix analysis of resonances in (n,n).
8262.4 4	5/2 ⁺		G	J^π : L(n,n)=2 and R-Matrix analysis of resonances in (n,n).
8273.3 4	1/2		G	J^π : R-Matrix analysis of resonances in (n,n).
8298.27 30			G	
8333.8 4	3/2 ⁻		G	J^π : L(n,n)=1 and R-Matrix analysis of resonances in (n,n).
8335.9 4			G	
8391.3 4	3/2 ⁻		G	J^π : L(n,n)=1 and R-Matrix analysis of resonances in (n,n).
8393.64 30			G	
8406.55 30			G k	XREF: k(8430)
8418.8 4	(1/2 to 9/2) ⁺		G k	XREF: k(8430) J^π : L(p, ^3He)=2.
8755.7 13	(19/2)	<1 ps	E	$T_{1/2}$: from ($^{18}\text{O}, 2\alpha n\gamma$) in 2021Go09 with residual Doppler energy shifts observed for the 732 γ and 1576 γ peaks.
9155 10	(1/2 to 9/2) ⁺		K	$T=5/2$ J^π : L(p, ^3He)=2.
10048.4 10	(19/2)		E	
12469.1 15	(21/2)		E	

[†] From E_γ data when measured γ -ray energies are available. In other cases weighted averages are taken of values available from different reactions.

[‡] From analyzing power measurement or/and the comparison of the DWBA prediction of cross section with experimental data in particle-transfer reactions or/and from angular distribution measurements of γ rays. When assigning J^π to a level based on γ transitions from that level to a level with known J^π , evaluators follow rules: if $E_\gamma < 4$ MeV, only M1, E1, or E2 are considered possible; if $E_\gamma > 4$ MeV, M2 and E3 are also considered possible. In particle transfer reactions, target $J^\pi=0^+$ for ^{34}S in (d,p) reaction and $J^\pi=3/2^+$ for ^{37}Cl in (p, ^3He) and (d, α) reactions.

[#] From (d,p γ) and ($^{32}\text{S}, n\gamma$) using DSAM (Doppler shift attenuation method). Weighted averages taken when values from both are available.

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Adopted Levels, Gammas (continued)

 ^{35}S Levels (continued)

@ Band(A): $\Delta J=1$ sequence based on $7/2^-$.

& Band(B): Sequence based on $3/2^+$.

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{S})$								
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	$\delta^\#$	Comments
1572.378	1/2 ⁺	1572.327 12	100	0.0	3/2 ⁺	(M1)		E_γ : weighted average of 1572.252 29 from $^{35}\text{P } \beta^-$ decay, 1576 2 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{S}\gamma)$, 1572.4 3 from $(^{14}\text{N}, 3p\gamma)$, 1572.333 8 from (n, γ) E=thermal, 1572.24 17 from $(d, p\gamma)$, and 1572 1 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{S}\gamma)$. Mult.: D from $(^{14}\text{N}, 3p\gamma)$; $\Delta\pi$ =no from level scheme.
1991.28	7/2 ⁻	1991.26 5	100	0.0	3/2 ⁺	M2+E3	+0.17 6	E_γ : weighted average of 1990 2 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{S}\gamma)$, 1991.3 2 from $(^{14}\text{N}, 3p\gamma)$, 1990.5 4 from $(^{18}\text{O}, 2\alpha n\gamma)$, 1991.27 5 from (n, γ) E=thermal, and 1991 1 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{S}\gamma)$. Mult., δ : from $(d, p\gamma)$. D+Q or Q+O from $\gamma(\theta)$ in $(d, p\gamma)$; E2+M3 is ruled out by RUL; D+Q is inconsistent with level scheme from 7/2 ⁻ to 3/2 ⁺ .
2347.789	3/2 ⁻	356.66 [@] 9 775.398 6	<0.084 [@] 35.3 27	1991.28 1572.378	7/2 ⁻ 1/2 ⁺	[E2] (E1)		E_γ, I_γ : from (n, γ) E=thermal. E_γ : from (n, γ) E=thermal. Others: 778 2 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{S}\gamma)$ and 775.4 4 from $(^{14}\text{N}, 3p\gamma)$. I_γ : weighted average of 37 5 from $(^{14}\text{N}, 3p\gamma)$, 35 4 from (n, γ) E=thermal, 37.0 28 from $(d, p\gamma)$, and 33.3 27 from $(d, \alpha\gamma)$. Mult.: D(+Q) from $\gamma(\theta)$ in $(d, p\gamma)$; $\Delta\pi$ =yes from level scheme; M2 ruled out by RUL.
		2347.700 19	100.0 27	0.0	3/2 ⁺	(E1)		E_γ : weighted average of 2352 2 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{S}\gamma)$, 2347.8 3 from $(^{14}\text{N}, 3p\gamma)$, 2347.700 15 from (n, γ) E=thermal, and 2347 2 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{S}\gamma)$. I_γ : from $(d, \alpha\gamma)$. Others: 100 8 from $(^{14}\text{N}, 3p\gamma)$, 100 8 from (n, γ) E=thermal, and 100.0 28 from $(d, p\gamma)$. Mult.: D(+Q) from $\gamma(\theta)$ in $(d, p\gamma)$; $\Delta\pi$ =yes from level scheme; M2 ruled out by RUL.
2717.00	5/2 ⁺	368.7 5	14 8	2347.789	3/2 ⁻	[E1]		E_γ : weighted average of 368.5 4 from (n, γ) E=thermal and 370 1 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{S}\gamma)$. I_γ : unweighted average of 6.7 23 from (n, γ) E=thermal and 22 4 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{S}\gamma)$.
		725.1 1145.7 2717.01 16	<5 <5 100 10	1991.28 1572.378 0.0	7/2 ⁻ 1/2 ⁺ 3/2 ⁺	[E1] [E2] M1+E2	-0.39 5	I_γ : from $(d, p\gamma)$. I_γ : from $(d, p\gamma)$. E_γ : weighted average of 2720 2 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{S}\gamma)$, 2717.0 4 from $(^{14}\text{N}, 3p\gamma)$, 2716.99 16 from (n, γ) E=thermal, and 2717 2 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{S}\gamma)$. I_γ : from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{S}\gamma)$. Other: 100 17 from (n, γ) E=thermal. Mult., δ : D+Q from $\gamma(\theta)$ in $(d, p\gamma)$ with $\delta=-0.39$ 5 or -15.0 34 for $J_i=5/2$ (1975VaYG); -15.0 34 is unlikely based on RUL; E1+M2 is ruled out by RUL.
2938.64	(3/2) ⁺	586 942 1363	<10 <20 <15	2347.789 1991.28 1572.378	3/2 ⁻ 7/2 ⁻ 1/2 ⁺			I_γ : from $(d, p\gamma)$. I_γ : from $(d, p\gamma)$. I_γ : from $(d, p\gamma)$.

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{S})$ (continued)							Comments
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	
2938.64	(3/2) ⁺	2938.57 18	100	0.0	3/2 ⁺		E_γ : weighted average of 2938.29 40 from ^{35}P β^- decay, 2943 2 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{S}\gamma)$, and 2938.58 11 from (n, γ) E=thermal.
3421.1	5/2 ⁺	486	<4	2938.64	(3/2) ⁺	[M1,E2]	I_γ : from (d,p γ).
		703.0	<6	2717.00	5/2 ⁺	[M1,E2]	I_γ : from (d,p γ).
		1072.2	<7	2347.789	3/2 ⁻	[E1]	I_γ : from (d,p γ).
		1428.1	<30	1991.28	7/2 ⁻	[E1]	I_γ : from (d,p γ).
		1848.6	<7	1572.378	1/2 ⁺	[E2]	I_γ : from (d,p γ).
		3423 2	100	0.0	3/2 ⁺	[M1,E2]	E_γ : weighted average of 3425 2 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{S}\gamma)$ and 3421 2 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{S}\gamma)$.
3558.09	(3/2 ⁻ , 5/2 ⁺)	619.23 19	8.9 17	2938.64	(3/2) ⁺		I_γ : from (d,p γ).
		1210.28 4	54 6	2347.789	3/2 ⁻		E_γ, I_γ : from (n, γ) E=thermal.
							E_γ : from (n, γ) E=thermal. Other: 1211 2 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{S}\gamma)$.
3594.8	3/2 ⁺ , 5/2 ⁺	1566.7 3	100 6	1991.28	7/2 ⁻		I_γ : weighted average of 54 6 from (n, γ) E=thermal and 54 6 from (d, $\alpha\gamma$). Other: 18 7 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{S}\gamma)$.
							E_γ : from (n, γ) E=thermal. Other: 1567 2 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{S}\gamma)$.
							I_γ : from (d, $\alpha\gamma$). Others: 100 19 from (n, γ) E=thermal and 100 10 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{S}\gamma)$.
		1986 ^{&} 2	39 6	1572.378	1/2 ⁺		E_γ, I_γ : from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{S}\gamma)$.
		3558.1 5	18 4	0.0	3/2 ⁺		E_γ, I_γ : from (n, γ) E=thermal.
		657	<10	2938.64	(3/2) ⁺		I_γ : from (d,p γ).
		877.6 5	9.3 23	2717.00	5/2 ⁺		E_γ : from ($^{14}\text{N}, 3\text{p}\gamma$).
							I_γ : from ($^{14}\text{N}, 3\text{p}\gamma$). Other: <11 from (d,p γ).
		1243	<15	2347.789	3/2 ⁻		I_γ : from (d,p γ).
		1599	<17	1991.28	7/2 ⁻		I_γ : from (d,p γ).
3801.958	3/2 ⁻	2020	<17	1572.378	1/2 ⁺		I_γ : from (d,p γ).
		3594.5 7	100 9	0.0	3/2 ⁺	Q	E_γ : weighted average of 3594.7 7 from ($^{14}\text{N}, 3\text{p}\gamma$), 3594.4 11 from ($^{18}\text{O}, 2\alpha\text{n}\gamma$), and 3593 2 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{S}\gamma)$. Other: 3602 2 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{S}\gamma)$.
							$I_\gamma, \text{Mult.}$: from ($^{14}\text{N}, 3\text{p}\gamma$).
		863.28 28	1.3 4	2938.64	(3/2) ⁺	[E1]	E_γ : from (n, γ) E=thermal.
							I_γ : from (n, γ) E=thermal. Other: <8.9 from (d,p γ).
		1084.79 15	2.2 7	2717.00	5/2 ⁺	[E1]	E_γ : from (n, γ) E=thermal.
							I_γ : weighted average of 2.1 4 from (n, γ) E=thermal and 11 5 from (d,p γ).
		1454.09 4	18 4	2347.789	3/2 ⁻	[M1,E2]	E_γ : from (n, γ) E=thermal.
							I_γ : weighted average of 18 4 from (n, γ) E=thermal and 18 7 from (d,p γ).
		1809.1	<8.9	1991.28	7/2 ⁻	[E2]	I_γ : from (d,p γ).
		2229.510 16	88 15	1572.378	1/2 ⁺	[E1]	B(E1)(W.u.)=0.0010 8
							E_γ : from (n, γ) E=thermal.
							I_γ : unweighted average of 110 11 from (n, γ) E=thermal, 93 9 from (d,p γ), and 61 5 from (d, $\alpha\gamma$).

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{S})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. #	$\delta^\#$	Comments
3801.958	$3/2^-$	3801.73 4	100 5	0.0	$3/2^+$	[E1]		E_γ : from (n, γ) E=thermal. I_γ : from (d, $\alpha\gamma$). Other: 100 9 from (d,py), 100 9 from (n, γ) E=thermal.
3818.1	$5/2^-, 7/2^-$	1824.7 3	100	1991.28	$7/2^-$	M1+E2	+0.55 9	E_γ : weighted average of 1824.6 2 from (¹⁴ N,3py), 1824.4 6 from (¹⁸ O,2 $\alpha\gamma$), and 1827 1 from ²⁰⁸ Pb(³⁶ S, ³⁵ S γ). Other: 1830 2 from ⁹ Be(³⁶ S, ³⁵ S γ). Mult., δ : from $\gamma(\theta)$, $\gamma\gamma(\theta)$ (ADO), and RUL in (¹⁴ N,3py). I_γ : from (d,py). I_γ : from (d,py). I_γ : from (d,py). E_γ : from (¹⁴ N,3py). Other: 1542 2 from ⁹ Be(³⁶ S, ³⁵ S γ). I_γ : unweighted average of 24 10 from (¹⁴ N,3py) and 89 9 from (d, $\alpha\gamma$). Other: <18 from (d,py). E_γ ,Mult.: from (¹⁴ N,3py). I_γ : from (d, $\alpha\gamma$). Other: 100 14 from (¹⁴ N,3py) and 100 from (d,py). I_γ : from (d,py). E_γ , I_γ : from ⁹ Be(³⁶ S, ³⁵ S γ). I_γ : from (d, $\alpha\gamma$). Other: <16 from (d,py).
3885.3	$(3/2^-, 5/2^+)$	464.0 950 1167.0 1538.4 8	<30 <12 <15 57 33	3421.1 2938.64 2717.00 2347.789	$5/2^+$ $(3/2^+)^+$ $5/2^+$ $3/2^-$			
		1894.9 4	100 11	1991.28	$7/2^-$	D		
3895	$1/2^+$	2312.6 1542 2 3884.8	<16 100 33 7	1572.378 2347.789 0.0	$1/2^+$ $3/2^-$ $3/2^+$			
4022.3	$11/2^-$	207.0 5 2032.0 3	0.59 30 100 6	3818.1 1991.28	$5/2^-, 7/2^-$ $7/2^-$	E2		E_γ : weighted average of 2031.8 3 from (¹⁴ N,3py) and 2032.4 4 from (¹⁸ O,2 $\alpha\gamma$). Other: 2032 2 from ²⁰⁸ Pb(³⁶ S, ³⁵ S γ). I_γ ,Mult.: from (¹⁴ N,3py). I_γ : from (d,py). I_γ : from (d,py). I_γ : weighted average of 86 22 from (d,py) and 97 12 from (d, $\alpha\gamma$). E_γ : from ⁹ Be(³⁶ S, ³⁵ S γ). I_γ : from (d,py). I_γ : from (d, $\alpha\gamma$). Other: 92 33 from (d,py). E_γ : from ⁹ Be(³⁶ S, ³⁵ S γ). I_γ : from (d,py). I_γ : from (d, $\alpha\gamma$). Other: 100 33 from (d,py). I_γ : from (d,py).
4025.9	$3/2^+, 5/2^+$	467 604.0 1088.5 1309 2	<14 <28 95 12 <22	3558.09 3421.1 2938.64 2717.00	$(3/2^-, 5/2^+)$ $5/2^+$ $(3/2^+)^+$ $5/2^+$			
		1679.5 2037 2	97 12 <28	2347.789 1991.28	$3/2^-$ $7/2^-$			
4106.4	$(5/2^+)$	2455.4 4024.8 549 686 1172 1389 1758 2114 2535	100 18 <36 <4.6 <4.6 15 7 <12 <5.8 <8.1 <5.8	1572.378 0.0 3558.09 3421.1 2938.64 2717.00 2347.789 1991.28 1572.378	$1/2^+$ $3/2^+$ $(3/2^-, 5/2^+)$ $5/2^+$ $(3/2^+)^+$ $5/2^+$ $3/2^-$ $7/2^-$ $1/2^+$	[M1,E2] [M1,E2] [M1,E2] [E1] [E1] [E2]		

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{S})$ (continued)							
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. #	Comments
4106.4	(5/2) ⁺	4105.3 8	100 7	0.0	3/2 ⁺	[M1,E2]	E_γ : from (n, γ) E=thermal. I_γ : from (d,p γ).
4180	(1/2,3/2,5/2) ⁺	2608	22 6	1572.378	1/2 ⁺		E_γ, I_γ : from (d, $\alpha\gamma$).
		4180	100 6	0.0	3/2 ⁺		I_γ : from (d, $\alpha\gamma$). Other: 100 10 from (d,p γ). E_γ : from (d, $\alpha\gamma$).
4189.280	1/2 ⁻	387	<7.7	3801.958	3/2 ⁻	[M1,E2]	
		597	<7.7	3594.8	3/2 ⁺ ,5/2 ⁺	[E1]	
		631.32 @ 24	<2.8 @	3558.09	(3/2 ⁻ ,5/2 ⁺)		E_γ, I_γ : from (n, γ) E=thermal.
		768	<7.7	3421.1	5/2 ⁺	[M2]	
		1250.61 5	8.4 10	2938.64	(3/2) ⁺	[E1]	E_γ, I_γ : from (n, γ) E=thermal.
		1471	<9.6	2717.00	5/2 ⁺	[M2]	
		1840.2 12	91 10	2347.789	3/2 ⁻	[M1,E2]	I_γ : weighted average of 87 16 from (n, γ) E=thermal and 92 10 from (d,p γ). E_γ : from (n, γ) E=thermal.
		2196	<15	1991.28	7/2 ⁻	[M3]	
		2616.8 13	9.5 28	1572.378	1/2 ⁺	[E1]	E_γ, I_γ : from (n, γ) E=thermal.
		4188.95 5	100 10	0.0	3/2 ⁺	[E1]	I_γ : from (d,p γ). Other: 100 11 from (n, γ) E=thermal. E_γ : from (n, γ) E=thermal.
4302.6	(1/2 to 7/2) ⁻	1954	100 9	2347.789	3/2 ⁻		
		4302	70 9	0.0	3/2 ⁺		
4477.64	(1/2,3/2,5/2) ⁺	595	<8.2	3885.3	(3/2 ⁻ ,5/2 ⁺)		
		678	<8.2	3801.958	3/2 ⁻	[E1]	
		888	<8.2	3594.8	3/2 ⁺ ,5/2 ⁺	[M1,E2]	
		922	<8.2	3558.09	(3/2 ⁻ ,5/2 ⁺)		
		1059	<16	3421.1	5/2 ⁺	[E2]	
		1545	<12				
		1760.55 11	96 10	2717.00	5/2 ⁺	[E2]	I_γ : weighted average of 100 15 from (n, γ) E=thermal, 100 12 from (d,p γ), and 92 10 from (d, $\alpha\gamma$). E_γ : from (n, γ) E=thermal.
		2131	<13	2347.789	3/2 ⁻	[E1]	
		2489.9	39 10				E_γ, I_γ : from (d, $\alpha\gamma$).
		2905.1 4	86 12	1572.378	1/2 ⁺	[M1,E2]	I_γ : weighted average of 96 28 from (n, γ) E=thermal, 64 12 from (d,p γ), and 100 10 from (d, $\alpha\gamma$). E_γ : from (n, γ) E=thermal.
		4479.7	26 5	0.0	3/2 ⁺	[M1,E2]	E_γ, I_γ : from (d, $\alpha\gamma$).
4839	(1/2 to 9/2) ⁺	1006.8 2	66 15	3818.1	5/2 ⁻ ,7/2 ⁻	D	E_γ : from (¹⁴ N,3p γ). Other: 1007.2 11 from (¹⁸ O,2 α n γ). I_γ : weighted average of 73 9 from (¹⁴ N,3p γ) and 36 18 from (¹⁸ O,2 α n γ). Mult.: from (¹⁴ N,3p γ) and (¹⁸ O,2 α n γ).
		1228.1 3	100 9	3594.8	3/2 ⁺ ,5/2 ⁺	D	E_γ : from (¹⁴ N,3p γ). Other: 1228.3 6 from (¹⁸ O,2 α n γ).

Adopted Levels, Gammas (continued)

<u>$\gamma(^{35}\text{S})$ (continued)</u>							Comments
<u>$E_i(\text{level})$</u>	<u>J_i^π</u>	<u>$E_\gamma^\dagger$</u>	<u>$I_\gamma^\ddagger$</u>	<u>E_f</u>	<u>J_f^π</u>	<u>Mult.[#]</u>	
							E_γ : from ($^{14}\text{N}, 3p\gamma$). Other: 1228.3 6 from ($^{18}\text{O}, 2\alpha\gamma$). I_γ : from ($^{14}\text{N}, 3p\gamma$). Other: 100 18 from ($^{18}\text{O}, 2\alpha\gamma$). Mult.: from ($^{14}\text{N}, 3p\gamma$).
4839	(1/2 to 9/2) ⁺	2831.4 9	36 9	1991.28	7/2 ⁻		
4899.7	9/2 ⁺	1305.3 4	55 18	3594.8	3/2 ⁺ , 5/2 ⁺		E_γ : from ($^{18}\text{O}, 2\alpha\gamma$). Other: 1304.8 12 from ($^{14}\text{N}, 3p\gamma$). I_γ : from ($^{14}\text{N}, 3p\gamma$). Mult.: from ($^{14}\text{N}, 3p\gamma$).
4903.32	1/2 ⁻	2908.3 5 1101.92 31 1964.8 2 2184.16 19 2555.492 14	100 18 0.43 11 4.9 13 62 7 41 4	1991.28 3801.958 2938.64 2347.789	7/2 ⁻ 3/2 ⁻ (3/2) ⁺ 3/2 ⁻	D	
4963.09	3/2 ⁻	3330.84 13 4902.99 5 1161.05 20 1404.967 24 2615.2 2 2972.0 4 3390.62 7 4963.06 22	100 9 53 5 0.98 14 10.7 11 19.5 18 3.2 11 100 9 47 5	1572.378 0.0 3801.958 3558.09 2347.789 1991.28 1572.378 0.0	1/2 ⁺ 3/2 ⁺ 3/2 ⁻ (3/2 ⁻ , 5/2 ⁺) 3/2 ⁻ 7/2 ⁻ 1/2 ⁺ 3/2 ⁺		
5010.2	11/2 ⁻	986.8 3	100 5	4022.3	11/2 ⁻	D	E_γ : weighted average of 986.9 3 from ($^{14}\text{N}, 3p\gamma$) and 986.6 3 from ($^{18}\text{O}, 2\alpha\gamma$). I_γ : from ($^{14}\text{N}, 3p\gamma$). Mult.: from ($^{18}\text{O}, 2\alpha\gamma$) and ($^{14}\text{N}, 3p\gamma$).
5412.2	9/2 ⁺	3018.5 7 1389.0 4	43 5 100 10	1991.28 4022.3	7/2 ⁻ 11/2 ⁻	E2 D	Mult.: from ($^{14}\text{N}, 3p\gamma$). E_γ : from ($^{14}\text{N}, 3p\gamma$). Other: 1389.0 8 from ($^{18}\text{O}, 2\alpha\gamma$). I_γ , Mult.: from ($^{14}\text{N}, 3p\gamma$).
5752.5	(1/2 to 9/2 ⁻)	1525.7 8 3420.6 9 5752.0 8	15.4 26 28.2 26 100	3885.3 1991.28 0.0	(3/2 ⁻ , 5/2 ⁺) 7/2 ⁻ 3/2 ⁺		
5841.485		4268.65 14	100	1572.378	1/2 ⁺		
5877.6	11/2 ⁺	465.4 2	33.8 26	5412.2	9/2 ⁺	D	E_γ : weighted average of 465.3 3 from ($^{14}\text{N}, 3p\gamma$) and 465.5 2 from ($^{18}\text{O}, 2\alpha\gamma$). I_γ : weighted average of 31.6 26 from ($^{14}\text{N}, 3p\gamma$) and 35.9 26 from ($^{18}\text{O}, 2\alpha\gamma$). Mult.: from ($^{18}\text{O}, 2\alpha\gamma$).
		867.6 4	50.0 26	5010.2	11/2 ⁻	D	E_γ : weighted average of 867.3 3 from ($^{14}\text{N}, 3p\gamma$) and 868.2 4 from ($^{18}\text{O}, 2\alpha\gamma$). I_γ : from ($^{14}\text{N}, 3p\gamma$). Other: 51 5 from ($^{18}\text{O}, 2\alpha\gamma$). Mult.: from ($^{18}\text{O}, 2\alpha\gamma$).

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{S})$ (continued)							Comments
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	
5877.6	11/2 ⁺	977.9 3	43 4	4899.7	9/2 ⁺	D+Q	E_γ : weighted average of 977.8 3 from ($^{14}\text{N},3\text{p}\gamma$) and 978.4 10 from ($^{18}\text{O},2\alpha\text{p}\gamma$). I_γ : weighted average of 42.1 26 from ($^{14}\text{N},3\text{p}\gamma$) and 67 18 from ($^{18}\text{O},2\alpha\text{p}\gamma$). Mult.: from ($^{14}\text{N},3\text{p}\gamma$).
		1055.1 6	80 5	4839	(1/2 to 9/2) ⁺	D	E_γ : unweighted average of 1054.5 3 from ($^{14}\text{N},3\text{p}\gamma$) and 1055.6 4 from ($^{18}\text{O},2\alpha\text{p}\gamma$). I_γ : weighted average of 79 5 from ($^{14}\text{N},3\text{p}\gamma$) and 100 26 from ($^{18}\text{O},2\alpha\text{p}\gamma$). Mult.: from ($^{18}\text{O},2\alpha\text{p}\gamma$).
		1854.4 4	44.7 26	4022.3	11/2 ⁻		E_γ : from ($^{14}\text{N},3\text{p}\gamma$). Other: 1854.5 42 from ($^{18}\text{O},2\alpha\text{p}\gamma$). I_γ : from ($^{14}\text{N},3\text{p}\gamma$). Other: 7E1 4 from ($^{18}\text{O},2\alpha\text{p}\gamma$).
		2061.6 4	98 5	3818.1	5/2 ⁻ , 7/2 ⁻	D	E_γ : weighted average of 2061.5 3 from ($^{14}\text{N},3\text{p}\gamma$) and 2063.3 14 from ($^{18}\text{O},2\alpha\text{p}\gamma$). I_γ : weighted average of 100 5 from ($^{14}\text{N},3\text{p}\gamma$) and 82 16 from ($^{18}\text{O},2\alpha\text{p}\gamma$). Mult.: from ($^{18}\text{O},2\alpha\text{p}\gamma$).
		2283.0 4	52 16	3594.8	3/2 ⁺ , 5/2 ⁺		E_γ : weighted average of 2282.9 4 from ($^{14}\text{N},3\text{p}\gamma$) and 2283.5 10 from ($^{18}\text{O},2\alpha\text{p}\gamma$). I_γ : unweighted average of 68.4 26 from ($^{14}\text{N},3\text{p}\gamma$) and 36 8 from ($^{18}\text{O},2\alpha\text{p}\gamma$).
6018.8	(1/2 to 9/2 ⁻)	3886.0 13	7.9 26	1991.28	7/2 ⁻		
6078.48	1/2 ⁻ , 3/2 ⁻	6018.2 6	100	0.0	3/2 ⁺		
		3139.9 5	22 7	2938.64	(3/2) ⁺		
		6077.87 11	100 12	0.0	3/2 ⁺		
6293.93	(1/2 to 9/2 ⁻)	6293.2 4	100	0.0	3/2 ⁺		
6299.4	11/2 ⁺	887.0 7	40 10	5412.2	9/2 ⁺	D	
		2275.9 6	100 20	4022.3	11/2 ⁻		
		2483.6 8	90 20	3818.1	5/2 ⁻ , 7/2 ⁻		
6334		2329.5 9	100 20	4022.3	11/2 ⁻	D	
		2535.8 11	50 6	3801.958	3/2 ⁻	E2	
6354.89	(1/2 to 9/2 ⁻)	6355.0 6	100	0.0	3/2 ⁺		
6419.9	(1/2 to 9/2 ⁻)	6419.3 11	100	0.0	3/2 ⁺		
6629.43	(1/2 to 9/2 ⁻)	6628.5 6	100	0.0	3/2 ⁺		
6761.0	(1/2 to 9/2 ⁻)	6760.3 12	100	0.0	3/2 ⁺		
(6986.10)	1/2 ⁺	356.66 [@] 9	<0.073 [@]	6629.43	(1/2 to 9/2 ⁻)		
		631.32 [@] 24	<0.123 [@]	6354.89	(1/2 to 9/2 ⁻)		
		692.16 5	0.238 31	6293.93	(1/2 to 9/2 ⁻)		
		907.608 23	1.00 18	6078.48	1/2 ⁻ , 3/2 ⁻		

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{S})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	Comments
(6986.10)	1/2 ⁺	1144.591 20	0.98 11	5841.485			
		2022.954 9	19.6 18	4963.09	3/2 ⁻		
		2082.83 17	28.4 27	4903.32	1/2 ⁻		
		2508.39 8	0.68 7	4477.64	(1/2,3/2,5/2) ⁺		
		2796.74 4	8.8 7	4189.280	1/2 ⁻		
		3183.95 4	10.9 11	3801.958	3/2 ⁻		
		4638.14 14	100 9	2347.789	3/2 ⁻		
		6985.7 10	0.064 14	0.0	3/2 ⁺		
7179.8	15/2 ⁺	827.8 6	2.2 8	6334			
		880.2 4	5.9 8	6299.4	11/2 ⁺	[E2]	
		1302.0 3	100 6	5877.6	11/2 ⁺	E2	E_γ : weighted average of 1302.2 2 from ($^{14}\text{N},3p\gamma$) and 1301.6 3 from ($^{18}\text{O},2\alpha n\gamma$).
8023.7	17/2 ⁺	844.2 6	100	7179.8	15/2 ⁺	D	I_γ , Mult.: from ($^{14}\text{N},3p\gamma$). E_γ : weighted average of 843.9 8 from ($^{14}\text{N},3p\gamma$) and 844.3 6 from ($^{18}\text{O},2\alpha n\gamma$). Mult.: from ($^{18}\text{O},2\alpha n\gamma$).
8755.7	(19/2)	731.6 15	57 22	8023.7	17/2 ⁺	D	
		1576.3 14	100 22	7179.8	15/2 ⁺		
10048.4	(19/2)	2868.7 8	100	7179.8	15/2 ⁺	Q	
12469.1	(21/2)	2420.6 11	100	10048.4	(19/2)	D	

[†] Values with ΔE are primarily from (d,p γ), (d, $\alpha\gamma$) and (n, γ) (E=thermal). Weighted average taken when E_γ 's are available from both. Others are deduced from level-energy differences.

[‡] Primarily from (d, $\alpha\gamma$), (d,p γ) and (n, γ). Weighted average taken when I_γ 's are available from more than one reference, unless otherwise noted.

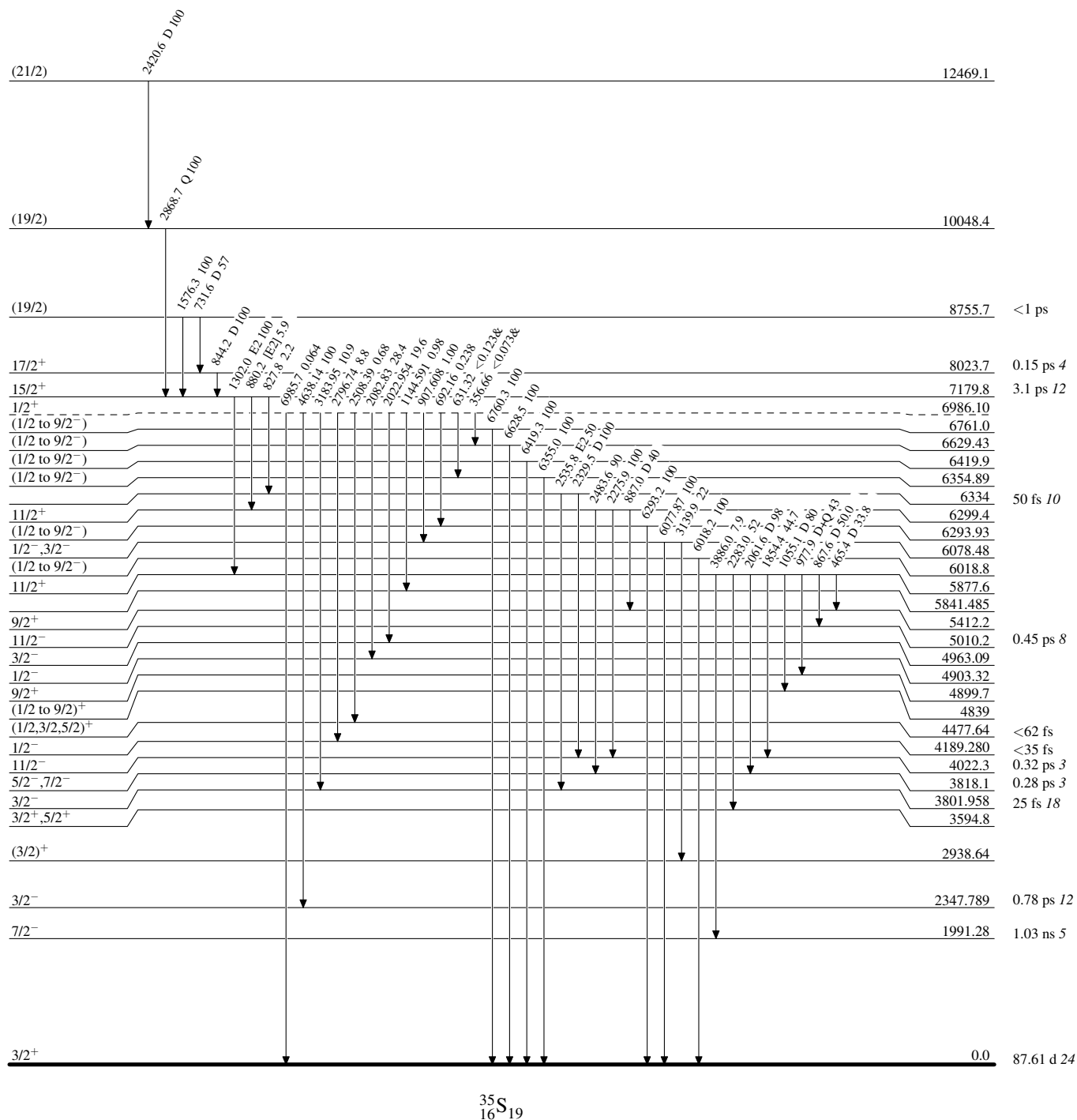
[#] From $\gamma(\theta)$ in (d,p γ). If $T_{1/2}$ is unknown and parity is determined not by polarization measurements, evaluators use D and Q, instead of M1 and E2, or, E1 and M2.

[@] Multiply placed with undivided intensity.

[&] Placement of transition in the level scheme is uncertain.

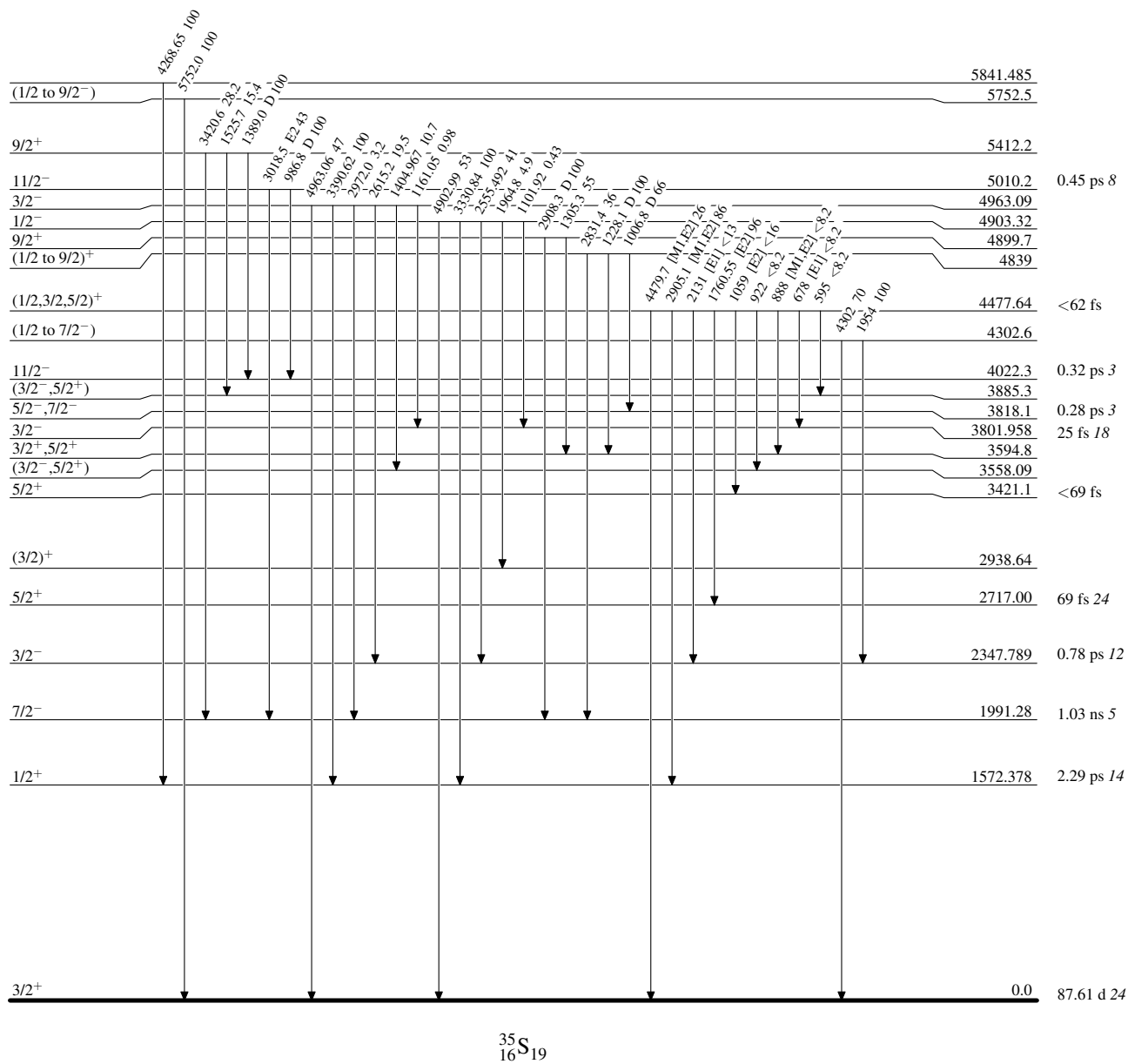
Adopted Levels, GammasLevel Scheme

Intensities: Relative photon branching from each level
& Multiplied: undivided intensity given



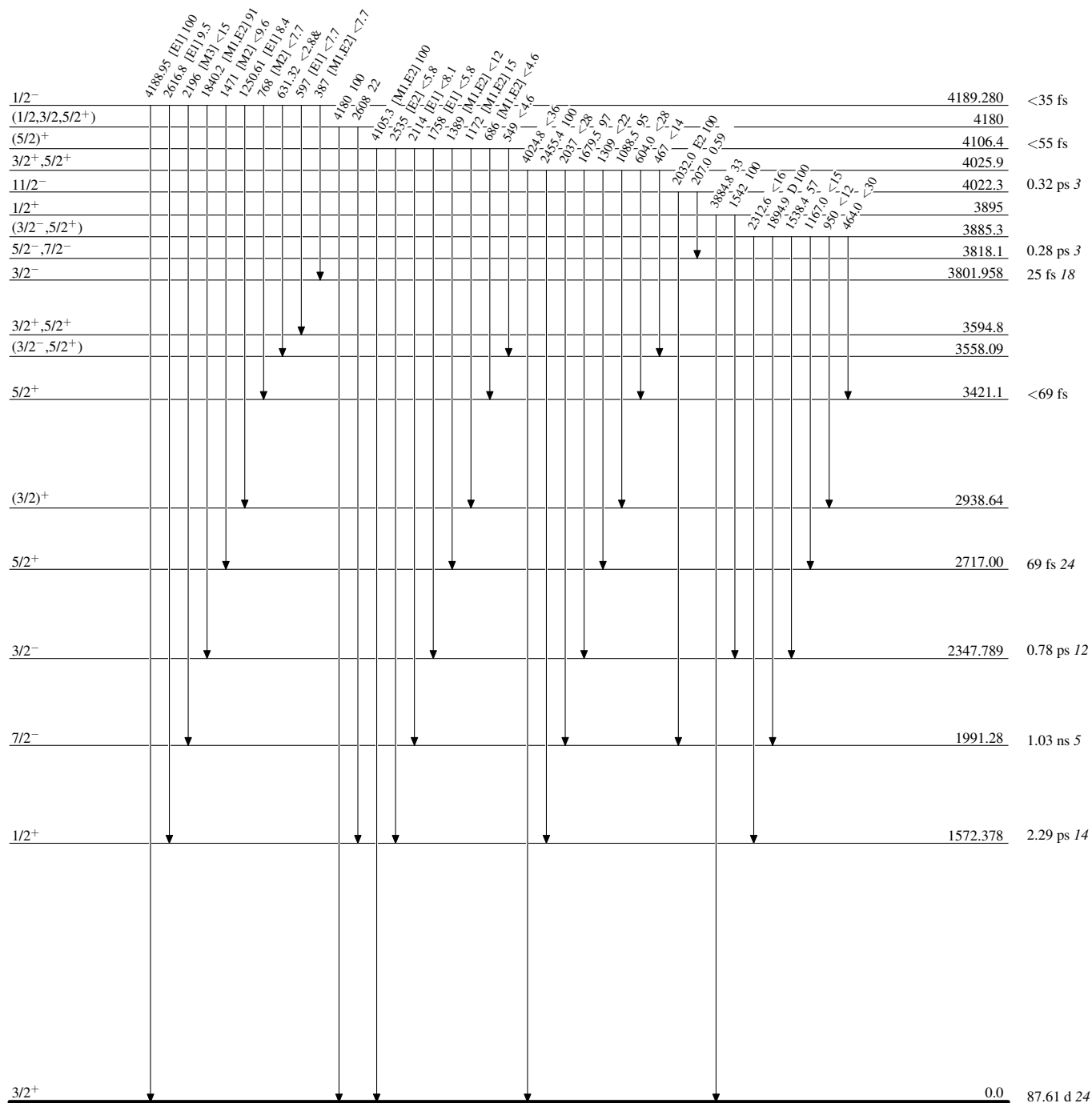
Adopted Levels, GammasLevel Scheme (continued)

Intensities: Relative photon branching from each level
& Multiply placed: undivided intensity given



Adopted Levels, GammasLevel Scheme (continued)

Intensities: Relative photon branching from each level
& Multiplicity placed: undivided intensity given



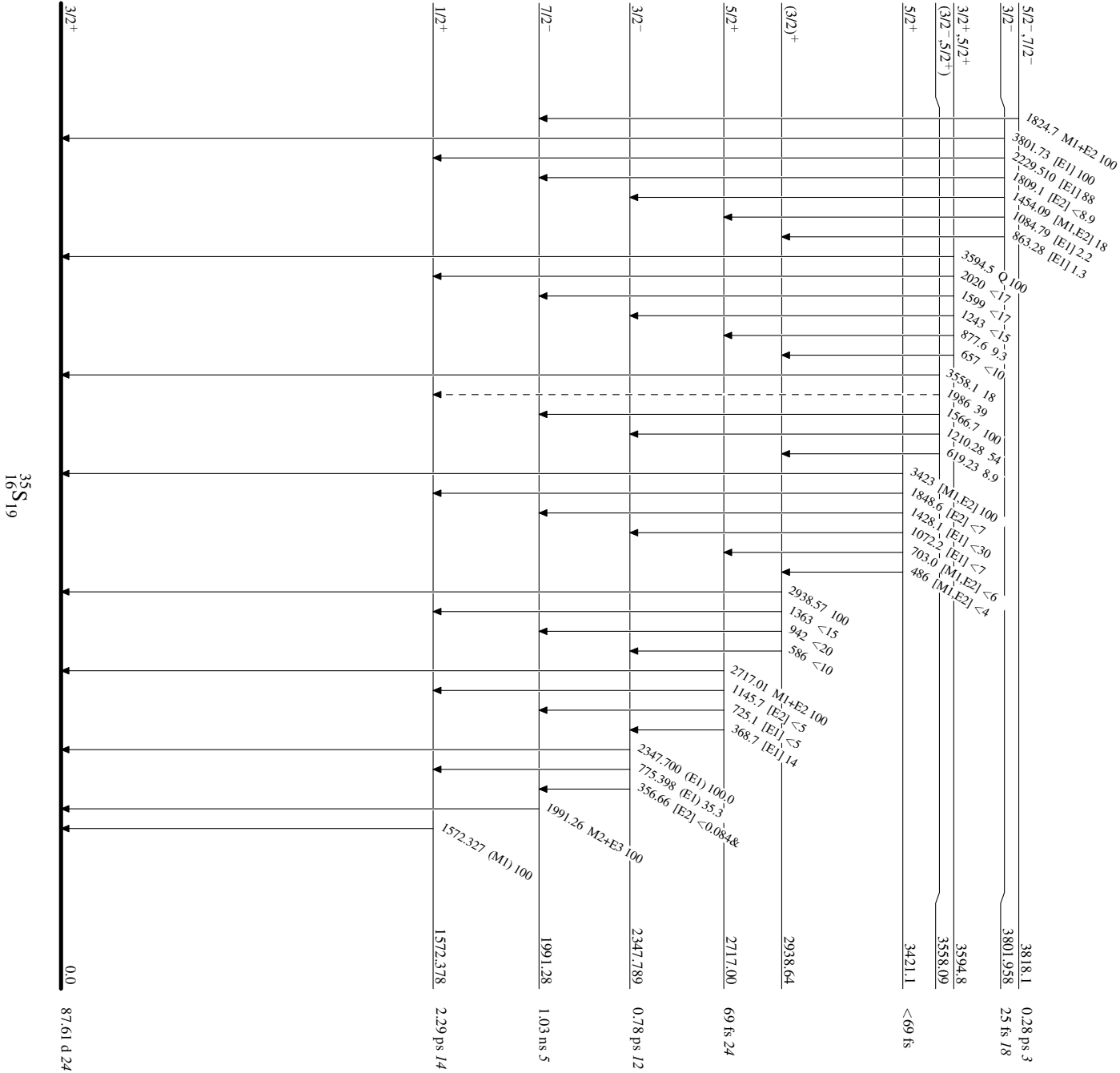
Adopted Levels, Gammas

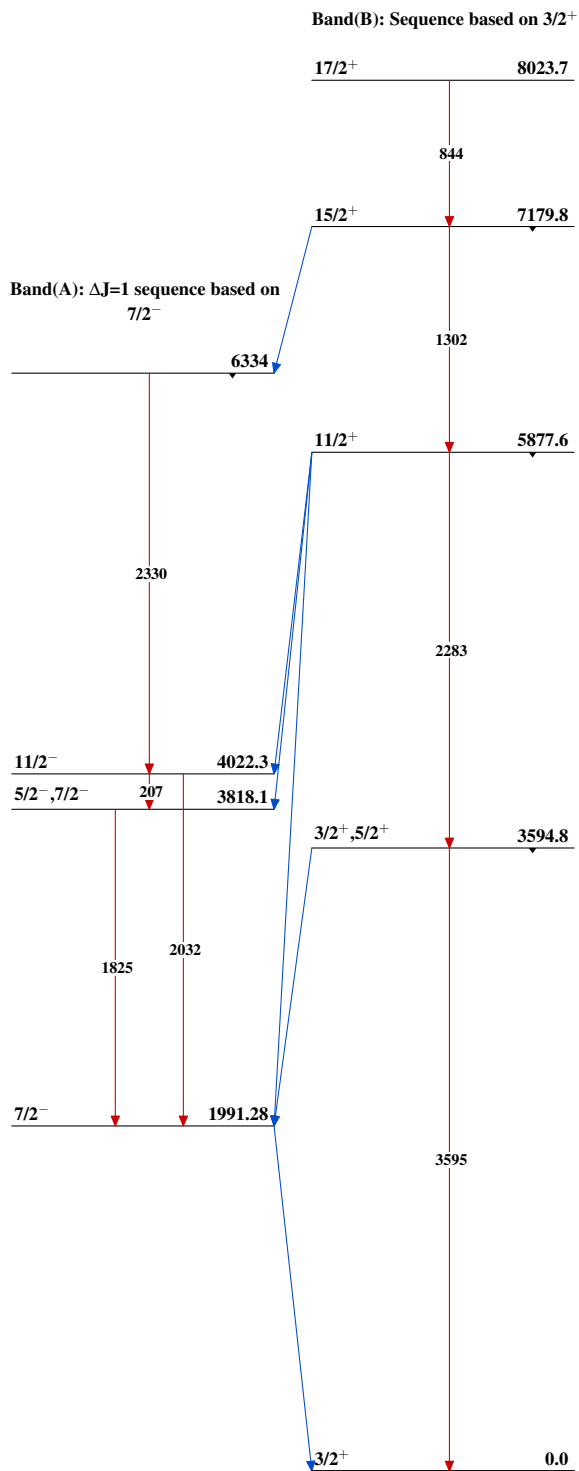
Level Scheme (continued)

Intensities: Relative photon branching from each level
& Multiply placed: undivided intensity given

Legend

-----▶ γ Decay (Uncertain)



Adopted Levels, Gammas $^{35}_{16}\text{S}_{19}$